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Data Resource Profile: Health Insurance Review and Assessment Service Korean Nationwide Claims OMOP-CDM (2015–2024) database, *"HIRA K-OMOP"*

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Abstract

The Health Insurance Review and Assessment Service Korean Nationwide Claims OMOP-CDM database (HIRA K-OMOP) is a nationwide data resource formatted according to the Observational Medical Outcomes Partnership (OMOP) Common Data Model (CDM) and derived from South Korea's National Health Insurance claims data. It includes patient-level information and insurance claims for the entire population of South Korea from 2015 to 2024, providing population-scale coverage (56,416,773 patients). The HIRA K-OMOP database captures comprehensive data generated under the National Health Insurance program, including diagnoses, medications (inpatient and outpatient prescriptions and pharmacy dispensing records), procedures, and claim costs. This enables high-resolution observation of healthcare utilization and clinical practice among the Korean population.

The database was constructed by mapping Korean Electronic Data Interchange (EDI) codes to OMOP standard concepts. The requisite mapping tables were developed by the investigators of the "Integrating Local Vocabulary into OMOP CDM" study and released in June 2025, allowing nationally generated claims data to be transformed into an analysis-ready research resource. The complete analytic vocabulary set underpinning the build was obtained from the official OHDSI ATHENA portal (athena.ohdsi.org) in March 2025. Key strengths of the database include deterministic linkage to the national death registry and the explicit separation of prescribing and dispensing records, which enhances the validity of drug exposure assessments. The database is accessible to researchers through a privacy-preserving distributed research environment managed by HIRA.

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Key Features

- The Health Insurance Review and Assessment Service Korean Nationwide Claims OMOP-CDM database (*HIRA K-OMOP*) is a nationwide data resource formatted according to the Observational Medical Outcomes Partnership (OMOP) Common Data Model (CDM) and derived from South Korea's national health insurance claims data.[1,2]

- It includes patient-level information and insurance claims for the entire population of South Korea from 2015 to 2024, providing population-scale coverage (56,416,773 patients).

- The *HIRA K-OMOP* database includes comprehensive data recorded as part of the National Health Insurance program, encompassing diagnosis, medications (inpatient, outpatient prescriptions, and dispensing records), procedure and claim costs. This allows for high-resolution observation of healthcare utilization and clinical practice among the Korean population.[3,4]

- It was constructed by mapping Korean Electronic Data Interchange (EDI) codes to OMOP standard concepts (concept IDs). The requisite mapping tables were developed by the investigators of the "Integrating Local Vocabulary into OMOP CDM" study and released in June 2025.[5] Consequently, nationally generated claims data have been transformed into an analysis-ready research resource. The complete analytic vocabulary set, which underpins the build, was obtained from the official OHDSI ATHENA portal (athena.ohdsi.org) in March 2025.[6]

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DATA RESOURCE BASICS

Role of HIRA in health system of South Korea

The health insurance system in the Republic of Korea is a mandatory social health insurance programme known as the National Health Insurance (NHI), which covers approximately 97% of citizens and residents.[7] The NHI reimburses most medical expenses incurred during patient care, including diagnostic tests, medical procedures, and prescribed medications. The Health Insurance Review and Assessment Service (HIRA) is a quasi-governmental agency that plays a central role in this system by reviewing the appropriateness of services, evaluating quality of care, and monitoring healthcare utilisation and costs to ensure that care remains both effective and cost-efficient.

HIRA K-OMOP Database

HIRA has access to a vast number of medical claims, making it an essential source of healthcare data in Korea. To promote data-driven research, HIRA has implemented an open data policy that allows claims data to be used for public purposes such as healthcare improvement and scientific research. Most requests are pre-processed by HIRA staff according to approved research protocols and then made available within a secure analytic environment, where researchers can access and analyses the data safely.

In 2023, HIRA developed a nationally representative COVID-19 database (*HIRA COVID-19 OMOP* database) covering approximately 10 million individuals and published its profile.[8] The Observational Medical Outcomes Partnership (OMOP) Common Data Model (CDM), maintained by the Observational Health Data Sciences and Informatics (OHDSI),

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enables international distributed data analysis by applying the same query to healthcare data from multiple institutions. Recently, HIRA expanded the OMOP architecture to a population-wide scale, thereby supporting national-level analyses and aligning the database with emerging healthcare research priorities.[9,10]

Separation of prescribing and dispensing in Korea

Since July 2000, South Korea has strictly enforced a system that separates prescribing from dispensing. Physicians are responsible for providing medical care and issuing prescriptions, while pharmacists are responsible for supplying and dispensing medications. Under this system, patients may choose not to have a prescription filled, and pharmacists are allowed to substitute prescribed medication with bioequivalent products. As a result, out-of-hospital prescription records issued by healthcare providers and dispensing records from local pharmacies are largely overlapping but are not fully identical. Because all pharmacies submit insurance claims for medication dispensing to HIRA, the agency can accurately capture nationwide dispensing data. The *HIRA K-OMOP* database is designed to provide both provider-level prescribing information and patient-centric medication dispensing records. To prevent duplication in analyses, out-of-hospital prescription records and dispensing records are maintained in separate schemas (Figure 1).

Population coverage

The *HIRA K-OMOP* database was constructed using claims data for the entire NHI-covered population. It contains comprehensive medical records for 56 million insured patients.

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Frequency of data collection

Claims data are collected continuously as part of routine NHI operations. The *HIRA K-OMOP* database is refreshed on a biennial basis, with additional updates performed as needed for specific research initiatives.

Privacy policy

The *HIRA K-OMOP* database has advantages in terms of personal information protection. The identification numbers of insurance subscribers managed by HIRA are replaced with serial numbers with matching digits when transformed into OMOP–CDM. Each subscriber's original identification number is replaced with a surrogate key generated via a one-way random sequence assignment, in which unique random numbers are allocated and the lookup table is subsequently destroyed. As a result, there is no way to reconstruct the original IDs, and it is impossible to identify specific individuals from the transformed database.

DATA COLLECTED

Data source

A schematic diagram of the data source is presented in Figure 1. The dataset was obtained from the data warehouse operated by HIRA, which includes national insurance claims that were completed through the official review process between January 2015 and December 2024, spanning a total of ten years. The HIRA claims database is organized into several core tables including Provider Claim Information (200 table), Pharmacy Claim Information (201 table),

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Treatment Details (300 table), Disease Information (400 table), Out-of-Hospital Prescription Details (530 table), Pharmacy Dispensing Details (301 table), and Healthcare Institution Status Information.

The claim specification table contains key administrative and clinical attributes for each claim, such as the claim key, patient surrogate key, healthcare institution key, primary and secondary disease codes, treatment start and end dates, duration of treatment, insurer liability, and visit details. The Treatment Details table records procedures, medications, and treatment materials with their respective codes, dosage amounts, and administration frequencies. The disease table is defined at the claim-statement level and records up to five diagnoses per claim—one principal and up to four secondary—aligned to the relevant clinical department. The Out-of-Hospital Prescription Details table captures dosage, administration frequency, and treatment duration for medications prescribed outside hospitals. The Pharmacy Claim Information table records pharmacy-submitted claims and classifies them as either extemporaneous compounding or prescription-based dispensing. The Pharmacy Dispensing Details table records charges for pharmacist services (e.g., basic dispensing fees and medication counselling fees). It also includes product-level records for medicines prepared extemporaneously or dispensed against an issued prescription. The Healthcare Institution Status Information table provides information on institution identifiers, regional locations, and institution types. Mortality data were linked from the national death registry maintained by Statistics Korea.

Extract, Transform, Load

The target population consisted of all patients whose insurance claims had been reviewed between 2015 and 2024. Based on this population, claims data from January 1, 2015, to

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December 31, 2024 were extracted. The extracted dataset included a wide range of key patient information, including demographics (sex, age, type of insurance), visit type (outpatient or inpatient care), medical history (diagnoses, procedures, and treatments), examination history, and prescription and dispensing records (such as the number of days prescribed and list of medications dispensed). The detailed specifications of these key variables are provided in Table 1.

As shown in Figure 1, the extracted claims data were then transformed into the OMOP CDM version 5.3, enabling meaningful clinical information to be represented in a standardized structure suitable for large-scale observational research.[11] This transformation involved the construction of several primary OMOP tables, such as PERSON, CONDITION_OCCURRENCE, DRUG_EXPOSURE and COST, which are widely used in real-world evidence studies.

For terminology standardization, each domain table utilized the concept IDs from the OMOP standardized vocabulary. The mapping of Korean claims codes to OMOP concepts was performed using the EDI-SNOMED and EDI-RxNorm vocabulary mapping dictionaries applied by HIRA in March 2025.

Data distribution

Table 2 presents the number of individuals and records included in 17 data tables, excluding vocabulary tables. In total, the *HIRA K-OMOP* database includes information on 56,416,773 insured persons. The CONDITION_OCCURRENCE table, which stores all diagnosis records for these individuals, contains 28,150,894,981 entries. The DRUG_EXPOSURE table includes 26,966,367,283 records of prescribed medications, while the DRUG_EXPOSURE_D

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(Dispensing) table contains 19,553,296,933 dispensing events. The PROCEDURE_OCCURRENCE table has the largest volume, with 16,281,745,905 documented procedures.

The baseline characteristics of the overall population are summarized in Table S1. Among all insured individuals, 49.86% were women. Figure 2 displays the age-sex distribution of the population, grouped in five-year age intervals. The overall mortality proportion recorded in the database was 5.51%.

Data Resource Use

HIRA has successfully completed two pilot programs to provide controlled access to OMOP CDM data. The first pilot, conducted from July 2022 to June 2023, utilized the *COVID-19 OMOP* database and supported more than 40 research projects under a distributed research model. The second pilot, carried out from September 2023 to December 2024, implemented the newly developed *HIRA K-OMOP* database and facilitated over 15 projects. The research outputs from these pilots included studies examining the associations between COVID-19 and specific health conditions, as well as nationwide evaluations of drug effectiveness and safety.[12,13]

Strengths and weaknesses

(i) Population-scale coverage with outcome ascertainment and international interoperability. The *HIRA K-OMOP* database encompasses virtually the entire population of South Korea across outpatient and inpatient settings, enabling large-scale

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pharmacoepidemiology and health-services research. Deterministic linkage to the national death registry facilitates robust mortality outcome ascertainment. Conversion to the OMOP Common Data Model (CDM) enhances reproducibility and cross-database comparability and thus supports multi-institutional and multinational research.[14–17]

(ii) Explicit separation of prescribing and dispensing to reduce drug-exposure misclassification. Korea's separation of prescribing and dispensing is operationalised by recording provider prescribing and pharmacy dispensing as distinct events, thereby mitigating double counting and improving the validity of drug-exposure definitions. Figure 3 compares dispensing-to-prescribing ratios across six clinically relevant drug classes. Dispensing-to-prescribing ratios were approximately 98% for SGLT2 inhibitors, around 82% for GLP-1 receptor agonists (with roughly 18 % of issued prescriptions not filled), and only about 39 % for benzodiazepines and Z-drugs. These patterns underscore the value of dispensing data within *HIRA K-OMOP* for capturing actual medication use and reducing exposure misclassification.

(iii) Privacy-preserving, governance-ready research operations. The database is accessed within a secure, distributed research environment. Direct identifiers are never exposed, and export of patient-, claim- or institution-level granular statistics is restricted. Together with IRB and HIRA approvals, these controls provide a governance framework that balances analytical utility with rigorous protection of personal information.[18]

However, the current data resource also has several limitations. To more fully reflect real-world clinical practice, additional data would be valuable, including non-reimbursed services, over-the-counter (OTC) medications, vaccines, and laboratory test results. Because HIRA cannot capture items that are not reimbursed by the NHI or OTC drugs sold directly at

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pharmacies without prescriptions, important clinical exposures may be missing. This gap has the potential to act as a major confounder in healthcare research. For example, the use of OTC medications such as acetaminophen or non-steroidal anti-inflammatory drugs (NSAIDs) cannot be accounted for when assessing drug-related risks. In addition, due to the inherent nature of the claims data, the *HIRA K-OMOP* database does not include laboratory results, which limits the ability to conduct studies that require laboratory values, such as biomarker-based analyses or research involving disease severity and treatment response.[19,20]

Nevertheless, such limitations can be effectively mitigated by utilizing the *HIRA K-OMOP* database in conjunction with hospital-based CDMs, such as the Federated E-health Big Data for Evidence Renovation Network (FEEDER-NET).[21] While FEEDER-NET comprises hospital Electronic Medical Records (EMR) containing detailed clinical data—including laboratory results and non-reimbursed services—it operates as a distributed network where patient linkage across institutions is not feasible. Therefore, a comparative analysis that leverages both the population-wide coverage of *HIRA K-OMOP* and the deep clinical granularity of hospital CDMs would offer a robust strategy for generating real-world evidence. Indeed, the analytic value of HIRA CDM has already been validated through its use in studies published in major journals, further underscoring its potential for high-impact research.[22,23]

Data Resource Access

The database is maintained by the Big Data Department of the Health Insurance Review and Assessment Service (HIRA), and inquiries regarding its use should be directed to cdm@hira.or.kr. Access for domestic researchers requires approval from an institutional review board as well as authorisation from HIRA.

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Ethics approval

The HIRA is a quasi-governmental organization authorized under the relevant laws of the Republic of Korea to process personal information. Since the construction of the database itself did not constitute human subjects research, it was exempt from review by an institutional bioethics committee. However, for any distributed research conducted by individual investigators, submission to and approval from an institutional review board is mandatory.

Author contributions

D H Yu designed the conversion logic of the database, performed the data conversion, and wrote the manuscript. S M Lim and H C Shin validated the converted data using the Data Quality Dashboard (DQD).[24,25] M Kim reviewed data security for the converted database. S C You and R W Park provided the vocabulary mapping sets. C S Kim supervised all processes. All authors reviewed the manuscript and provided critical revisions.

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Conflict of interest

None declared.

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Figure 1. Composition of source data for building *HIRA K-OMOP* database & Clinical domain composition of *HIRA K-OMOP* database

Figure 2. Population distribution by sex and 5-year-old age group

Figure 3. Comparison of Prescribing and Dispensing Outcomes for Three Specific Drug Classes

Table 1. Key information included in the *HIRA K-OMOP* database

Domain	Main information
Claim specification	Age, sex, insurance type, start date of care, outpatient/inpatient; basic information related to provider/pharmacy claims
Medical history	Medical records (drugs, treatment materials, etc.), examination records, medication records, cost information; detailed data generated during care episodes
Diagnosis	Primary and up to four secondary diagnostic codes, clinical department or provider/facility specialty; information related to disease diagnoses
Prescription	Medication prescription including dosage instructions, frequency, duration; prescription orders issued by healthcare providers
Dispensing	Dispensed products, dispensing date, pharmacy information; records of medicines dispensed at pharmacies against issued prescriptions
Care site	Region, type of medical institution, clinical departments; information on the facilities where care was delivered
Death	List of individuals with death records; date of death

HIRA, Health Insurance Review & Assessment Service; OMOP, Observational Medical Outcomes Partnership.

Table 2. Number of persons and records in the *HIRA K-OMOP* database

OMOP Tables	N of person	N of records
CARE SITE		134,510
CONDITION ERA		13,914,785,534
CONDITION OCCURRENCE		28,150,894,981

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COST	9,674,459,591
DEATH	3,106,139
DEVICE EXPOSURE	1,205,738,896
DRUG ERA	12,066,151,039
DRUG EXPOSURE	26,966,367,283
DRUG EXPOSURE (Dispensing)	19,553,296,933
MEASUREMENT	117,772,042
OBSERVATION	464,983,664
OBSERVATION PERIOD	56,416,773
PAYER PLAN PERIOD	56,416,773
PERSON	56,416,773
PROCEDURE OCCURRENCE	16,281,745,905
VISIT OCCURRENCE	9,674,459,591
VISIT OCCURRENCE (Pharmacy)	5,227,233,312
Total	**143,470,379,739**

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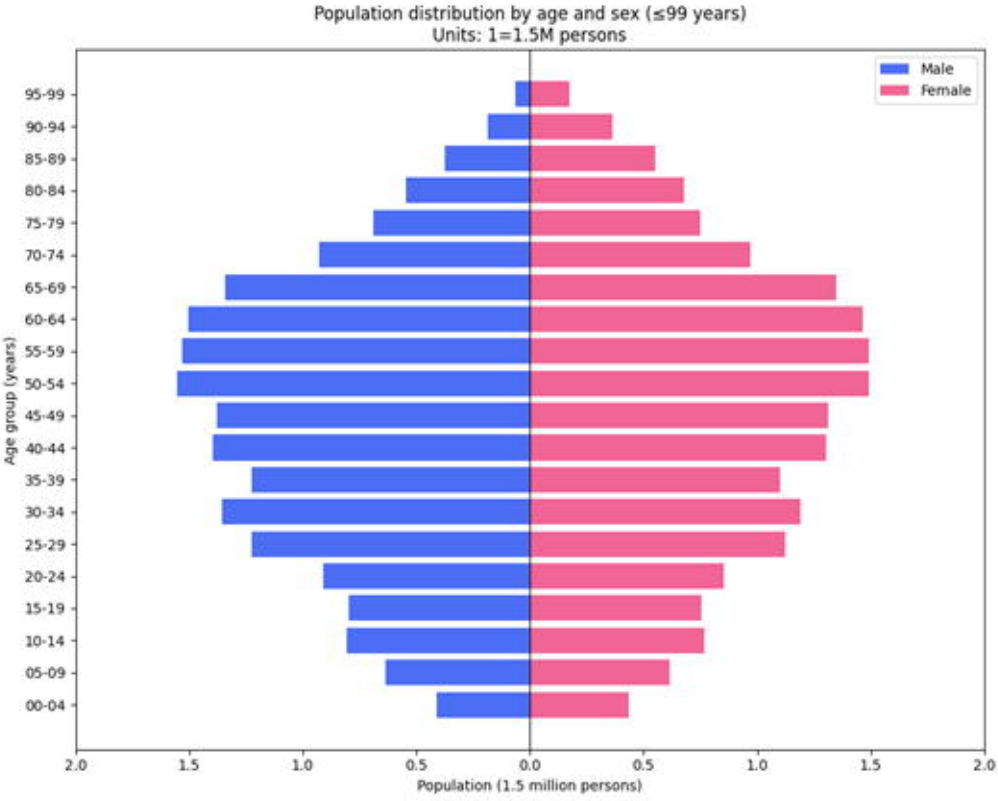
Population distribution by age and sex (≤99 years)
Units: 1=1.5M persons
Male
Female
95-99
90-94
85-89
80-84
75-79
70-74
65-69
60-64
55-59
50-54
45-49
40-44
35-39
30-34
25-29
20-24
15-19
10-14
05-09
00-04
Age group (years)
2.0
1.5
1.0
0.5
0.0
0.5
1.0
1.5
2.0
Population (1.5 million persons)

HIRA DW

Class Name	300T Injection/Infusion by provider	530T Outpatient-Prescription by provider	301T Dispensing by pharmacist	Dispensing Rate Against Prescriptions
SGLT-2	668,731	6,983,327	7,445,242	97.30%
GLP-1	335,823	1,451,049	1,455,520	81.46%
BZD/Z-drug	335,402,043	210,341,414	210,323,226	38.54%

K-OMOP

Class Name	Drug Exposure	Drug Exposure_D	Dispensing Rate Against Prescriptions
SGLT-2	7,551,536	7,444,109	98.58%
GLP-1	1,773,224	1,455,399	82.08%
BZD/Z-drug	566,868,789	221,593,412	39.09%
H2blocker	391,273,187	323,741,875	82.74%
PPI	213,185,996	184,910,566	86.74%
PCAB	30,930,988	26,970,911	87.20%

Windows 정품 인증
[설정]으로 이동하여 Windows를 정품

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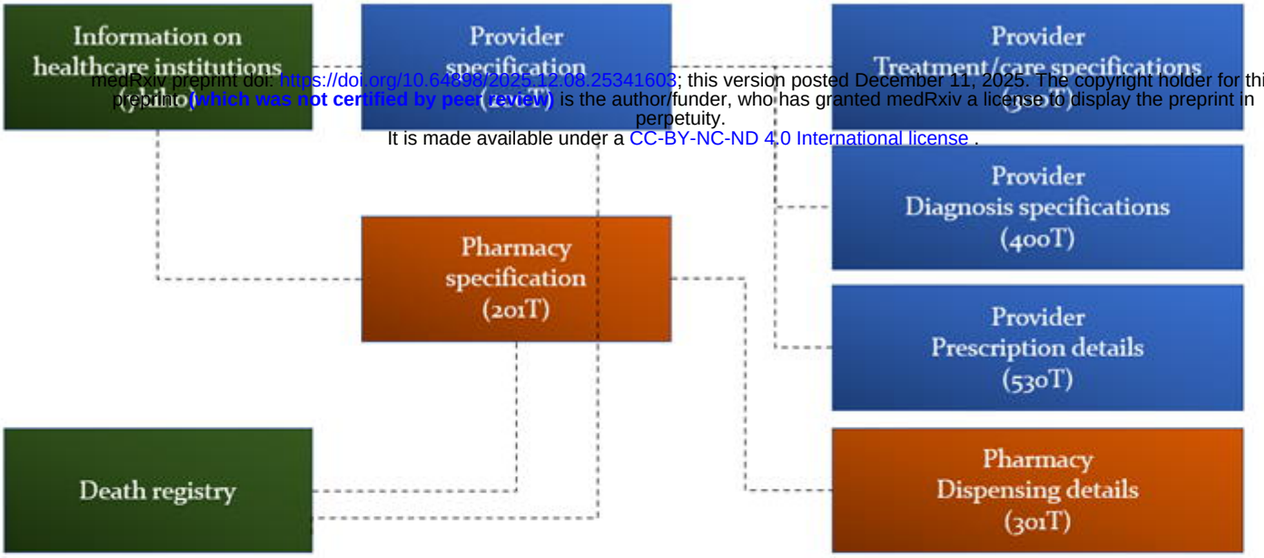
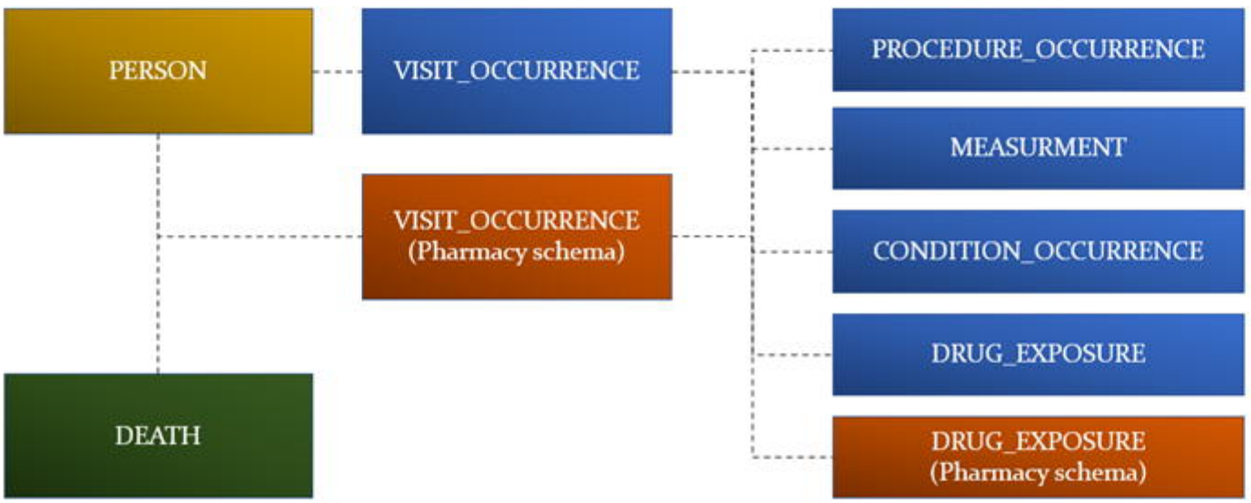
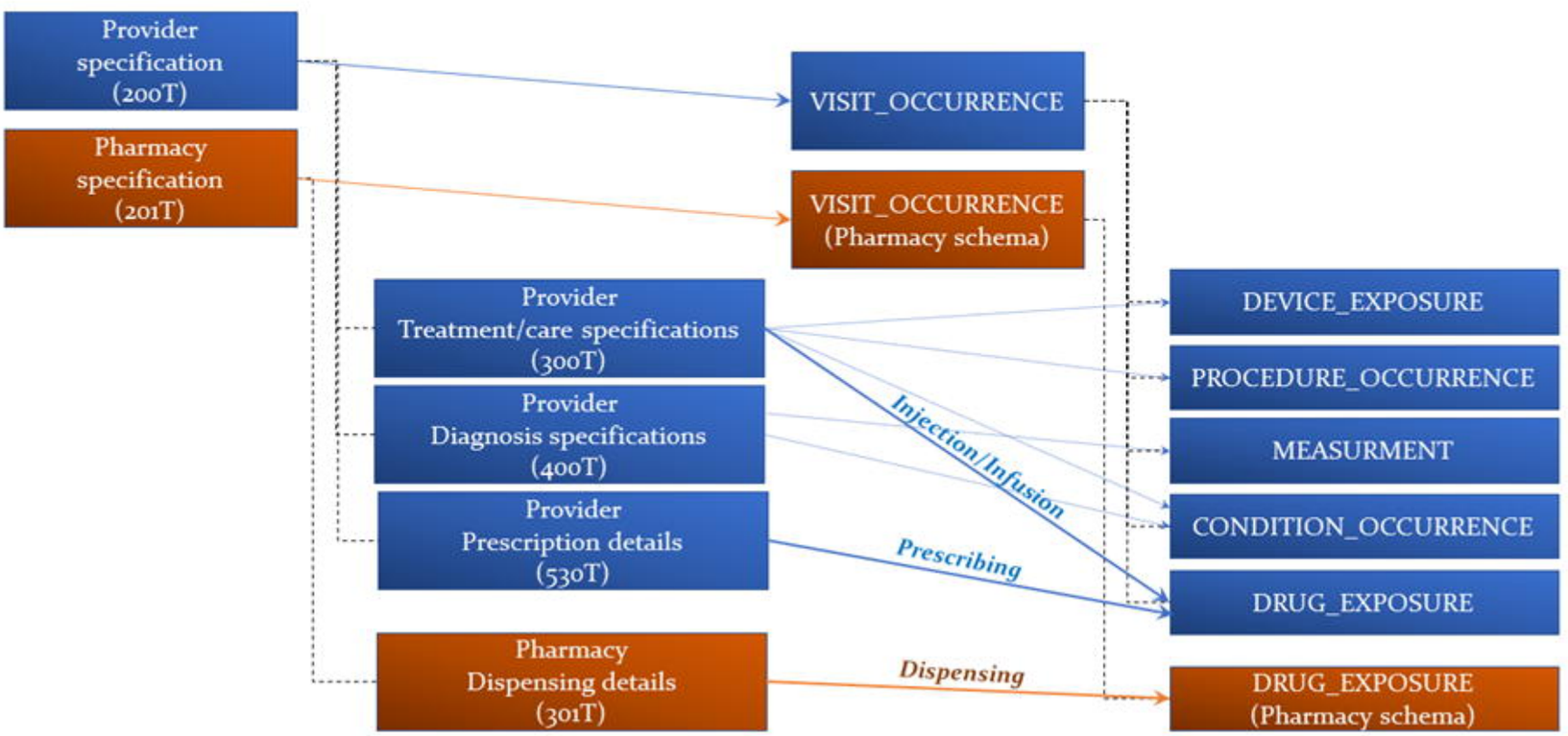
HIRA Data Warehouse



HIRA K-OMOP database (Clinical)



From HIRA DW To K-OMOP DB (Clinical)